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VIDEO INSIGHTS

Iberian killer whales assess deep-set fishing lines to detect hooked Atlantic bluefin tuna

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Abstract
A foraging-related behavior potentially allowing killer whales (*Orcinus orca*) to identify fishing lines with hooked Atlantic bluefin tuna (*Thunnus thynnus*) is described from the Strait of Gibraltar. On two days, four whales were video-recorded swimming repeatedly along the fishline, intermittently interacting with it by placing it on their backs or pressing, holding, or pushing it with the fins, snout, melon or belly. These interactions are consistent with assessing line tension or weight to detect the presence of a hooked fish at depth. Digital video images related to the article are available at http://www.momo-p.com/showdetail-e.php?movieid=momo260512oo01a and at http://www.momo-p.com/showdetail-e.php?movieid=momo260512oo02a.

Keywords Killer whale-fishery interactions · Bluefin tuna depredation · Line fishing behavior

Introduction

The Atlantic waters of the Iberian Peninsula host a population of 29–39 killer whales (*Orcinus orca*) associated with the movements of Atlantic bluefin tuna (ABFT), *Thunnus thynnus* (de Stephanis et al. 2008; Esteban et al. 2025). These killer whales temporarily inhabit the Strait of Gibraltar, intercepting the migratory route of ABFT on their way into the Mediterranean to spawn and on their return to the Atlantic feeding areas (Esteban et al. 2014; Sorell et al. 2017). During the post-spawning return migration (June–August), many ABFT congregate in a feeding area located approximately midway between Tangier, Morocco, and Tarifa, Spain (Fig. 1). This area, locally known as Las Bajas (The Shallows), is an elevated seabed plateau approximately 2 nmi long and 1 nmi wide, with a gradual slope descending from west to east and depths ranging from 290 to 500 m. During the six-hour rising tide, seafloor turbulence upwells nutrients that attract small prey species, which are then targeted by ABFT. Since 1995, fishermen from Spain and Morocco have come to this area to fish ABFT using a method called *stone fishing* (de la Serna and de Urbina 2010). Moroccan fishermen use 5–6 m long 15 hp boats. For fishing, they use a line with three or more hooks baited with octopus (*Octopus vulgaris*), squid (*Loligo vulgaris*) or mackerel (*Scomber scombrus*). The line has a 10–14 kg stone attached with a false knot. Once the stone reaches the bottom, the fishermen pull up the line and the false knot breaks, leaving the stone at the bottom. The baits then rise due to the continuous movement of the boat and the fishermen pulling the line by hand (Malouli Idrissi et al. 2013). The boat, moved by both tidal currents and the engine, keeps drifting westward toward the shallower part of Las Bajas. This movement keeps the baits near the seabed for longer, increasing fishing effectiveness, as ABFT are often hooked shortly after the stone reaches the bottom, suggesting that attacks occur at depth. Once ABFT is hooked, one or two fishermen lift the line by hand and bring the fish to the boat in a process that can take 20 min or more.

A group of the Iberian killer whales has learned to depredate on hooked ABFT by waiting near the surface and taking the fish in the first 2–50 m (Esteban et al. 2016a). Killer whales depredating on hooked fish of different species have been studied in other areas of the world (Guinet et al. 2015). Although this method carries risks associated with interactions with fishing gear (Lewison et al. 2004; Tixier et al. 2017, 2021; Herr et al. 2020), it is presumably more energetically efficient than chasing and capturing free-swimming ABFT, which involves prolonged high-speed pursuits until

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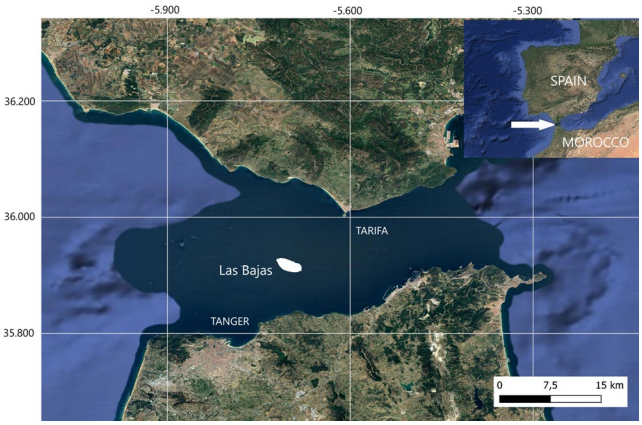


Fig. 1 Map of the Strait of Gibraltar showing Las Bajas fishing area (Google Earth, Google Inc., July 2025)

the ABFT becomes exhausted (Guinet et al. 2007; Esteban et al. 2016a). However, to depredate on ABFT, killer whales have to overcome a major problem: how to allocate their depredating time and energy. Fishing activity for returning migratory ABFT primarily occurs from June to August, and during this period, killer whales frequently visit Las Bajas to depredate on ABFT. The Moroccan fleet fishes in Las Bajas for one entire six-hour tidal cycle, starting with low tide each day (except on Fridays). On a typical fishing day, there are around 10–20 larger Spanish fishing boats and 50–60 small Moroccan fishing vessels in Las Bajas (personal observations; Malouli Idrissi et al. 2013). The data presented here refer to the Moroccan small vessels: total fishing time per stone throw—from throwing the stone to pulling the hooks out of the water when no ABFT is hooked—is on average 15 min (range 11–18 min; sample size $n=7$), and the average distance in a straight line covered by each vessel and the fishline is 0.89 km ($n=4$). During the entire 6-hour fishing period, from low to high tide, a single vessel throws an average of 18 stones (13–21; $n=10$), with one ABFT hooked for every ten stones deployed on average ($n=229$). This raises the question of how killer whales identify which fishing line holds a hooked tuna (Fig. 2).

Echolocation studies on Pacific killer whales preying on Chinook salmon (*Oncorhynchus tshawytscha*) have shown that prey detection ranges vary between 40 and 100 m, depending on acoustic conditions (Au et al. 2004). In Las Bajas—where depths range from 290 to 500 m—the high levels of noise produced by numerous fishing boats are likely to greatly reduce echolocation efficiency (Veirs et al. 2016). Theoretically, killer whales could rely on echolocation only by diving to considerable depths or by waiting

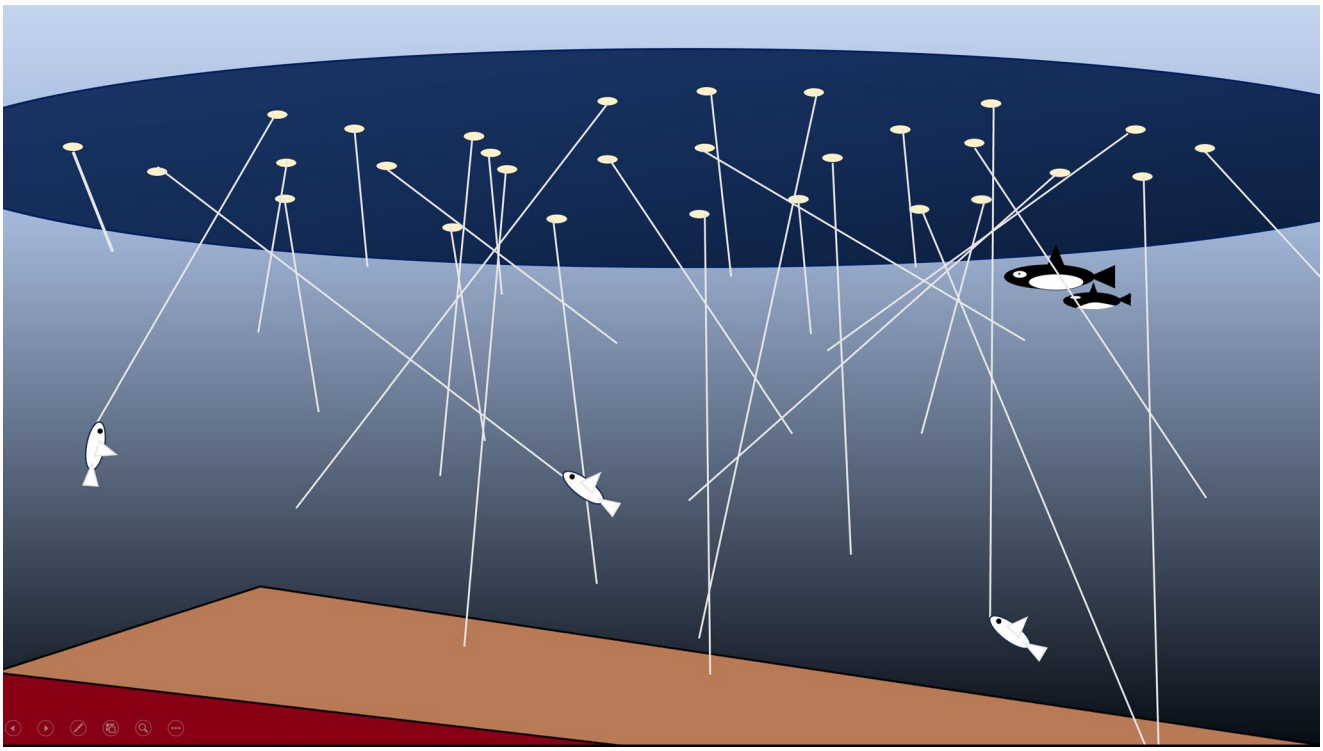


Fig. 2 Schematic representation (not to scale) of a scenario that may have led to the invention of line fishing behavior by Iberian killer whales. White dots on the surface represent fishing boats; only 29 boats are shown. Based on capture rates, only three lines carry hooked ABFT at a given time. A more realistic representation of a typical day, with 60 fishing lines actively fishing and considering that the total area of Las Bajas is approximately 6.9 km², would result in a density of about 8.7 lines per km², corresponding to an average spacing of approximately 340 m between lines

until the hooks ascend closer to the surface. Maximum dive depths of Iberian killer whales in Las Bajas remain unknown. Rudd et al. (2024) reported that typical dives by these killer whales in other parts of the Strait, where they actively pursue free-swimming ABFT, rarely exceed 100 m. Even if killer whales were capable of reaching depths near the seabed in Las Bajas, doing so for every encountered fishing line would likely be inefficient in terms of both energy expenditure and opportunity costs. The ability to assess whether a line carries a fish without such costs would thus represent a significant advantage. Fishermen frequently report a strong pull on the line during fishing events when killer whales are present, which they attribute to the killer whales physically interacting with the lines to assess whether a fish is hooked. During routine ABFT fishing prior to this study, we directly experienced this behavior on three occasions. In all of them, the tension on the line during killer whale interactions became so intense that the fisherman pulling the line was forced to release it. Although the hypothesis that killer whales interact with fishlines to assess whether they carry an ABFT is widely shared among local fishermen, it has not yet been scientifically verified. Here, underwater video recordings are used to evaluate the occurrence of this behavior and, if confirmed, to provide a detailed ethological description.

Video description

Fishing times in Las Bajas were measured using a smartphone chronometer application (Xiaomi Redmi Note 11). The distance travelled—from the moment the stone was thrown into the water until the hooks were retrieved onboard—was calculated by plotting the two reference points in Google Earth. The two video trials were conducted during regular fishing operations from a standard Moroccan fishing boat (with four people onboard) on August 12 and 17, 2023. A typical fishing line with three hooks and a stone attached to its end was used (Malouli Idrissi et al. 2013). A sports camera (GoPro Hero 11 Black, 4 K, 60 fps) was mounted on a 36-cm-long steel pole and slid along the fishing line using two carabiners welded to the pole. The camera, facing downward, was set to continuous recording mode and connected to an additional 60-m line to prevent further descent and potential damage due to pressure. DaVinci Resolve Studio version 20 was used to edit the footage from both video trials and to extract individual video frames. In both trials, killer whales were first sighted nearby, after which fishing began by setting the line with three baited hooks and a stone. Once the stone reached the seabed and the line began to be pulled up by hand, the camera was slid downward.

The first trial (Fig. 3; at http://www.momo-p.com/showdetail-e.php?movieid=momo260512oo01a) was conducted a few minutes before the onset of the rising tide, with only two other vessels operating nearby. The camera was deployed and reached a stable depth of 54 m within 45 s. Killer whales were first observed at 0:39 (minutes: seconds), with all four individuals ascending along the fishing line. They contacted the line with their pectoral fins and other parts of their bodies. From 0:57 to 1:29, the four killer whales passed the camera while ascending, possibly to breathe. From 3:36 to 4:18, they reappeared, swimming downward past the camera and continuing to follow the line approximately 20 m deeper. They remained in the area for a total of 4 min 18 s, circling and making contact with the line. From 7:54 to 8:57, all four killer whales were observed ascending past the camera once again, after which no further sightings occurred. In total, during the 9-min recording, the same group of four killer whales was recorded three times: twice ascending and once descending along the fishing line. The second trial (Fig. 4; http://www.momo-p.com/showdetail-e.php?movieid=momo260512oo02a) was conducted with the same group of four killer whales, near the end of the rising tide, when no vessels were present in the area. The camera descended for 29 s and remained at a depth of 36 m. The killer whales exhibited behavior similar to that observed in the first trial, periodically contacting the fishing line using various parts of their bodies. Unlike in the first trial, the four individuals did not appear to coordinate their movements. On several occasions, a single whale interacted with the line independently, and at one point, one individual ascending along the line crossed paths with another descending. The trial was interrupted due to the presence of floating buoys with lines, preventing the collection of quantitative data. Nevertheless, it is included here because it clearly demonstrates interactions with the fishing line involving the dorsal fin, belly, snout, and melon.

Discussion

Killer whales around the world exhibit a variety of foraging behaviors when depredating on different prey (Peterson et al. 2013; Guinet et al. 2015; Belonovich et al. 2021). Underwater video footage provides new insights into how Iberian killer whales may have learned to optimize depredation on ABFT by using fishing lines as reference paths to locate their prey at depth. The killer whales swim repeatedly along the line, intermittently interacting with it—likely assessing its weight and tension to determine whether it carries a fish. These interactions involve the snout, melon, belly, or fins (Fig. 3; at http://www.momo-p.com/showdetail-e.php?movieid=momo260512oo01a & Fig. 4; http://www.m

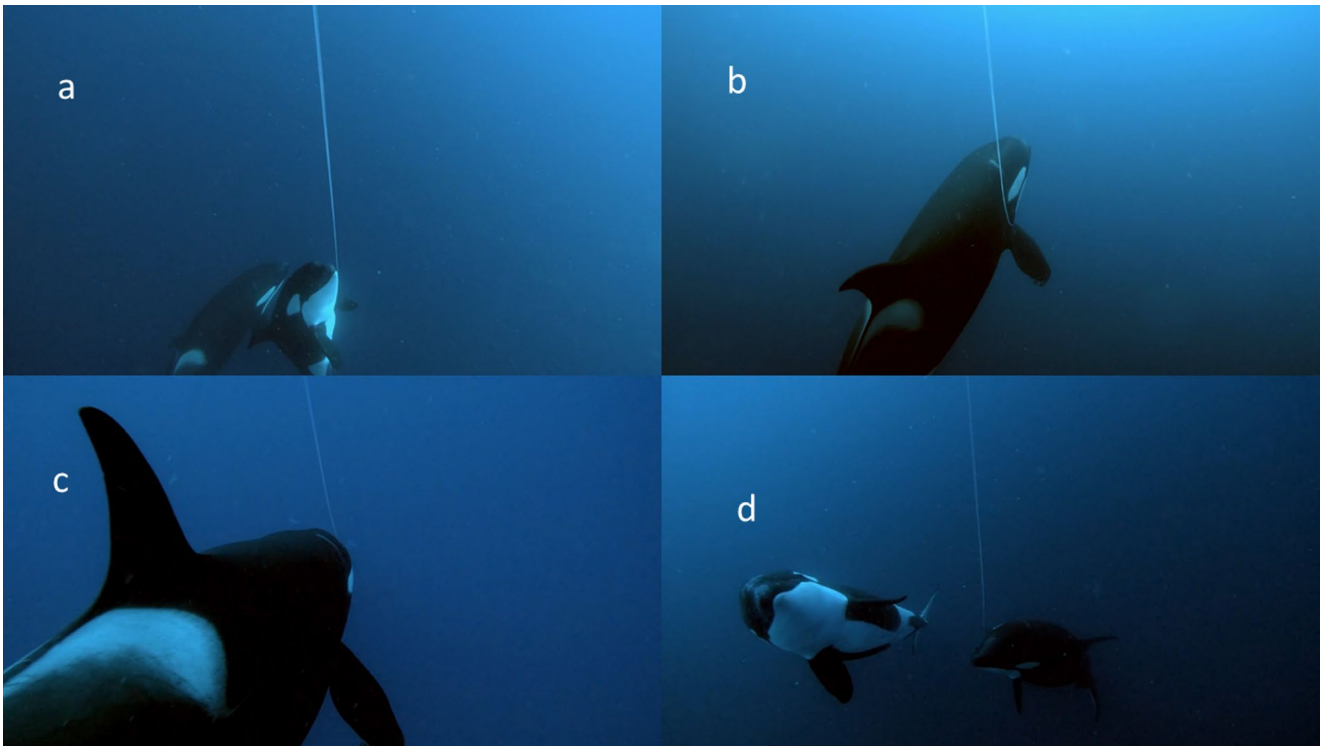


Fig. 3 Line fishing behavior recorded during the first trial during ABFT fishing operations in Las Bajas. The white vertical line visible in the images corresponds to a deep-set fishing line used during ABFT fishing operations. In all panels, the upper part of the image corresponds to the sea surface and the lower part to the seabed. (**a**) Two killer whales ascend alongside the fishing line. One individual appears to press its pectoral fin against the line while ascending. (**b**) A killer whale ascends while apparently holding the fishing line with the right pectoral fin. (**c**) Close-up view of a killer whale descending toward the fishing line while apparently pushing it with the head. (**d**) Two killer whales ascending alongside the fishing line, while one individual appears to maintain physical contact with it. These observations are consistent with tactile assessment of line tension or vibrations transmitted through the fishing line, possibly to detect hooked ABFT

omo-p.com/showdetail-e.php?movieid=momo260512oo02a). Killer whales may perform this behavior for variable periods before moving to another line. In the first trial, the killer whales remained at a line for 9 min, repeatedly testing it. This raises a question: if they interact with the line to detect hooked fish, why remain at a single line for such a long time? That day, we arrived at Las Bajas before low tide, marking the start of fishing activity. Only two other vessels were present, and it is unknown whether the killer whales had already visited them. The limited number of lines may have encouraged extended investigation. How long killer whales remain at a line likely depends on several factors, with fishing vessel density being particularly important. The two videos presented correspond to a camera fixed at a single position on the line; therefore, only interactions occurring in the immediate meters below the camera were recorded. However, additional interactions can be inferred from the strong pulling of the fishing line experienced by the fishermen onboard. Other indirect evidence of killer whale interactions with the lines includes sudden curvatures or abrupt movements of the line, which may indicate contacts beyond the camera field of view (Fig. 3; at http://www.momo-p.com/showdetail-e.php?movieid=momo260512oo01a). Distinctive marks on the killer whale skin may also reflect prior direct contact with the lines (Herr et al. 2020; Fig. 3; at http://www.momo-p.com/showdetail-e.php?movieid=momo260512oo01a).

Although the two video trials presented here do not provide statistically significant quantitative evidence for this foraging technique, the observations are consistent with interpreting line fishing as a culturally transmitted behavior. According to the criteria proposed by Whitehead and Rendell (2015), culture requires (1) behavioral variation and (2) social transmission. With respect to the first condition, not all Iberian killer whales depredate hooked ABFT (Esteban et al. 2016a), and this foraging strategy appears to be relatively recent, coinciding with the introduction of *stone fishing* in 1995. Regarding the second condition, although the origin of line fishing behavior is unknown, it is likely that one individual—either by accident or innovation—initiated the behavior, which was subsequently socially transmitted to others. The high fishing line density found in Las Bajas may have facilitated the spread or independent emergence of the line fishing behavior (Fig. 2). Observations from

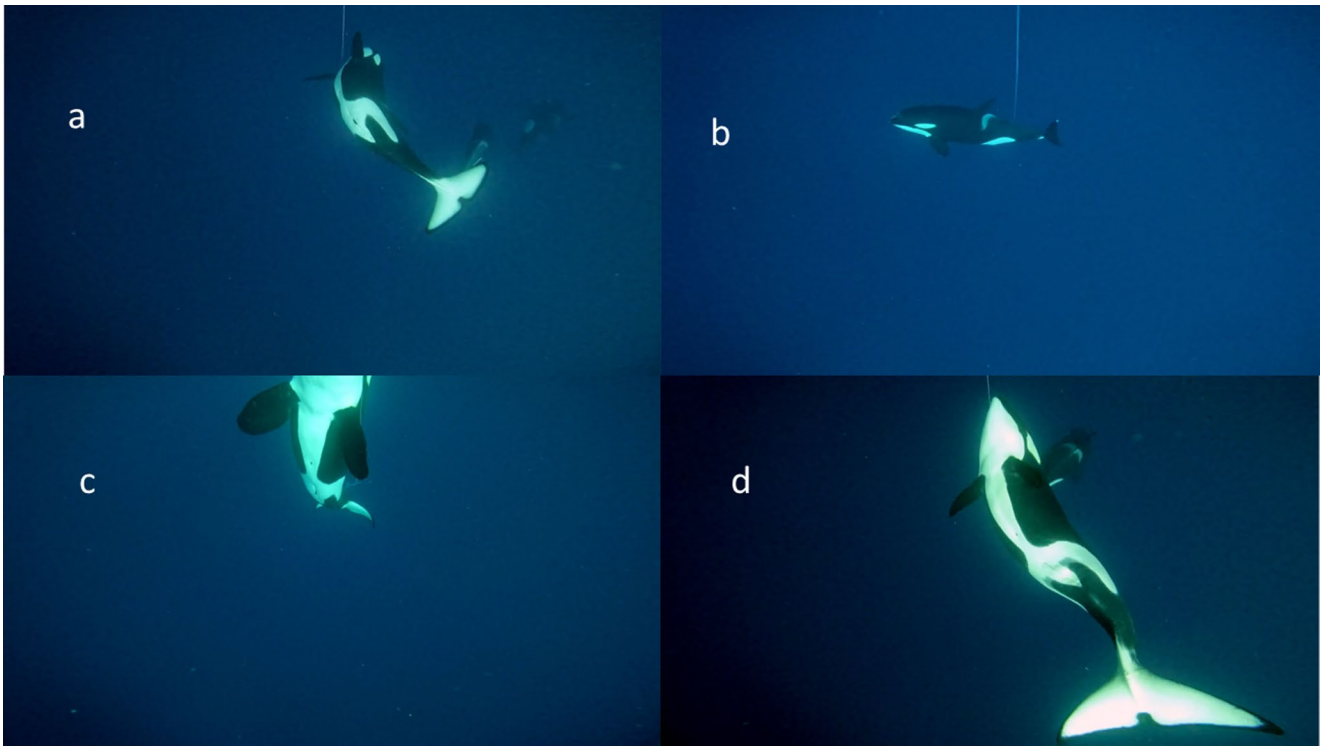


Fig. 4 Line fishing behavior recorded during the second trial during ABFT fishing operations in Las Bajas. The white vertical line visible in the images corresponds to a deep-set fishing line used during ABFT fishing operations. In all panels, the upper part of the image corresponds to the sea surface and the lower part to the seabed. (**a**) A killer whale descends immediately adjacent to the fishing line while apparently pressing the ventral body surface against it. (**b**) A killer whale pauses during ascent while maintaining the fishing line pressed against the ventral body surface. (**c**) Ventral view of a killer whale ascending while maintaining close body proximity to the fishing line. (**d**) Close-up view of a killer whale descending while apparently pressing the head against the fishing line. These observations are consistent with tactile assessment of line tension or vibrations transmitted through the fishing line, possibly to detect hooked ABFT

other killer whale populations indicate that fishery-related foraging behaviors can spread progressively through social learning among social units (Auguin et al. 2025), and Iberian killer whales depredating ABFT from fishlines form cohesive matrilineal pods, supporting the potential for vertical social transmission (Esteban et al. 2016b), although quantitative evidence remains limited. Future studies on these killer whales are necessary to explore further the line fishing behavior. A study comparing the presence of line marks on the bodies of depredating versus non-depredating killer whales would help confirm this behavior. The use of 'catcams'—small animal-borne video cameras mounted on free-ranging animals—would allow recording the continuous behavior of a whale on a single fishline as well as while moving to others (Rudd et al. 2024). In both trials presented here, no ABFT were captured. Observing killer whale behavior when they first detect a hooked ABFT would be of great interest. Do they emit high-amplitude vocalizations to alert the group? If so, how do the other killer whales respond? Do they act cooperatively to capture the fish from the hook?

Behavioral traits encoded in genes are passed from one generation to the next, but cultural behaviors can be transmitted every time the behavior is expressed—provided there is an observer (Whiten 2019; Whitehead et al. 2019). By swimming along the fishline and testing its tension from time to time, Iberian killer whales are effectively fishing themselves. This innovation constitutes a cultural behavior and is part of the *Orcinus orca* cultural patrimony. Killer whales in the Strait of Gibraltar are often perceived by fishermen as thieves stealing their catch. However, these killer whales were hunting ABFT in these waters long before human fishing began. It is likely that the unique conditions of Las Bajas have historically made it a traditional foraging ground for killer whales. From this perspective, it is we—humans—who are now taking their prey. The Iberian killer whale population is critically endangered due to prey depletion and habitat degradation (Esteban and Foote 2019). By inventing and adopting line fishing, these killer whales have demonstrated a remarkable capacity to adapt to the increasing anthropogenic exploitation of their prey (Esteban et al. 2016a; Tixier et al. 2017; Auguin et al. 2025). As suggested by Whitehead et al. (2004), non-human cultures should be

recognized and integrated into conservation strategies. It is our responsibility, beginning with science, to protect these killer whales and their unique cultural heritage for future generations.

What does the video tell us?

Killer whales prey on hooked ABFT by maintaining proximity to the fishline, swimming alongside it, and occasionally testing its weight using different parts of their bodies, possibly to determine whether a fish is hooked below.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s10164-026-00890-y.

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Declarations

Conflict of interest The author declare that he has no conflict of interest.

Ethical Approval This study did not involve any experimental procedures on animals. Observations were conducted in the natural habitat without any interference or manipulation. Therefore, ethical approval was not required.

Compliance with ethical standards All procedures performed in this study involving animals were in accordance with the ethical standards of the institution and/or national research committee and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards.

Informed Consent No human participants were involved in this study; thus, informed consent was not applicable.

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